DIABETMIN TABLET

VIDIA42-var3 (MY)

DESCRIPTION

Diabetmin 250mg Tablet
Round, white film-coated tablet, bevel-edged and shallow convex faces and scored on one side.

Diabetmin XR 500mg Tablet
Oblong, white uncoated tablet with deep convex faces and 'HD' embossed on one face.

COMPOSITION
Diabetmin 250mg Tablet
Metformin Hydrochloride 250mg per tablet.

Diabetmin XR 500mg Tablet
Metformin Hydrochloride 500mg per tablet (Extended-release)

PHARMACODYNAMICS

Metformin is an oral biguanide antidiabetic agent. Its mode of action is thought to be multifactorial and includes delayed uptake of glucose from the gastro-intestinal tract; increased peripheral glucose utilization mediated by increased insulin sensitivity; and inhibition of increased hepatic and renal gluconeogenesis.

PHARMACOKINETICS

Metformin Hydrochloride is slowly and incompletely absorbed from the gastrointestinal tract. The absolute bioavailability of a single 500mg dose is reported to be about 50 to 60%, although this is reduced somewhat if taken with food. Plasma protein binding is negligible. It is excreted unchanged in the urine. The plasma elimination half-life is reported to range from about 2 to 6 hours after oral administration.

INDICATIONS

Metformin is used in the treatment of non-insulin-dependent diabetes mellitus (type 2) in adults, not responding to exercise and dietary modification. Diabetmin may be used as monotherapy or in combination with other oral antidiabetic agents, or with insulin.

Note: For Extended Release Tablet: Diabetes mellitus who experienced gastrointestinal side effects with normal metformin.

CONTRAINDICATIONS

This medication is contraindicated in patients with the following medical problems:

- Hypersensitivity to metformin.
- Any condition needing close blood glucose control, such as: severe burns, dehydration, diabetic coma, diabetic ketoacidosis, hyperosmolar nonketotic coma, severe infection, major surgery, and severe trauma.
- Conditions associated with hypoxemia, such as: cardiorespiratory insufficiency, cardiovascular collapse, congestive heart failure, acute myocardial infarction.
- Severe, acute, or chronic hepatic disease.
- Active or history of lactic acidosis.
- Renal function impairment or renal disease.
- Diagnostic or medical examinations using intravascular iodinated contrast media such as: angiography, intravenous cholangiography, computed tomography (CT) scan, pyelography and urography.
- Severely reduced kidney function (GFR <30 mL/min).
- Any type of acute metabolic acidosis (such as lactic acidosis, diabetic ketoacidosis).

PRECAUTIONS

System Components and Performance: The polymers in Diabetmin XR 500 will hydrate and swell upon contact with water or gastrointestinal fluid. Metformin is released slowly from the dosage form by a process of diffusion through the gel matrix that is essentially independent of pH. The product may be excreted as a swelled mass in the stool.

Lactic acidosis: Lactic acidosis, a very rare but serious metabolic complication, most often occurs at acute worsening of renal function or cardiorespiratory illness or sepsis. Metformin accumulation occurs at acute worsening of renal function and increases the risk of lactic acidosis. In case of dehydration (severe diarrhoea or vomiting, fever or reduced fluid intake), metformin should be temporarily discontinued and contact with a health care professional is recommended. Medicinal products that can acutely impair renal function (such as antihypertensives, diuretics and NSAIDs) should be initiated with caution in metformin-treated patients. Other risk factors for lactic acidosis are excessive alcohol intake, hepatic insufficiency, inadequately controlled diabetes, ketosis, prolonged fasting and any conditions associated with hypoxia, as well as concomitant use of medicinal products that may cause lactic acidosis. Patients and/or care-givers should be informed of the risk of lactic acidosis. Lactic acidosis is characterised by acidotic dyspnoea, abdominal pain, muscle cramps, asthenia and hypothermia followed by coma. In case of suspected symptoms, the patient should stop taking metformin and seek immediate medical attention. Diagnostic laboratory findings are decreased blood pH (<7.35), increased plasma lactate levels (>5 mmol/L) and an increased anion gap and lactate/pyruvate ratio.

Renal impairment: GFR should be assessed before treatment initiation and regularly there after [See Section Recommended Dosage].

Metformin is contraindicated in patients with GFR <30 mL/min and should be temporarily discontinued in the presence of conditions that alter renal function [See Section Contraindications].

Surgery: Metformin should be discontinued before elective surgery with general anaesthesia and should not be usually resumed earlier than 48 hours afterwards.

Other precautions:

- All patients should continue their diet with a regular distribution of carbohydrate intake during the day. Overweight patients should continue their energy-restricted diet.
- The usual laboratory tests for diabetes monitoring should be performed regularly.
- Metformin alone never causes hypoglycaemia, although caution is advised when it is used in combination with insulin or sulfonylureas.

20216

20216

1214

DIABETMIN TABLET

Metformin may reduce vitamin B12 serum levels. The risk of low vitamin B12 levels increases with increasing metformin dose, treatment duration, and/or in patients with risk factors known to cause vitamin B12 deficiency. In case of suspicion of vitamin B12 deficiency (such as anemia or neuropathy), vitamin B12 serum levels should be monitored. Periodic vitamin B12 monitoring could be necessary in patients with risk factors for vitamin B12 deficiency. Metformin therapy should be continued for as long as it is tolerated and not contraindicated and appropriate corrective treatment for vitamin B12 deficiency provided in line with current clinical guidelines.

PREGNANCY AND LACTATION

For fertility and pregnancy problems, adequate and well-controlled studies in humans have not been done and documented. For patient plans to become pregnant or during pregnancy, control of blood glucose with diet alone or a combination of diet and insulin is recommended, while use of metformin is discouraged. Metformin is distributed into breast milk, but safety for use in nursing mothers has not been established.

MAIN SIDE/ADVERSE EFFECTS

Metformin can cause:

- Gastrointestinal adverse effects including anorexia, diarrhea, dyspepsia, flatulence, nausea, vomiting.
- Headache, metallic taste, weight loss.
- Anemia, megaloblastic, hypoglycemia, lactic acidosis.
- Long-term metformin therapy may cause a decrease of vitamin B12 absorption with decrease of serum levels.

Metabolism and nutrition disorders:
Common: Vitamin B12 decrease/deficiency.

DRUG INTERACTIONS

Concurrent use of this medication with the following may interact with metformin:

- Acute or chronic ingestion of alcohol.
- Cimetidine or other cationic medications excreted by renal tubular transport.
- Furosemide.
- Hyperglycemia-causing and hypoglycemia-causing medications.

OVERDOSE

Symptoms of overdose
Hypoglycemia and lactic acidosis.

Treatment of overdose

- For hypoglycemia
  Treating with immediate ingestion of a source of glucose and counseling patient to obtain emergency medical assistance immediately.
- For lactic acidosis:
  Hemodialysis with sodium bicarbonate.

DOSAGE AND ADMINISTRATION

Oral.

Monotherapy and combination with other oral antidiabetic agents:
Usual adult dose:
Diabetmin 250mg Tablet
2 to 3 times, 2 tablets (500mg) daily with meals.

Diabetmin XR 500mg Tablet
Initial dose of one tablet once daily with meals, maximum 4 tablets daily.

Note for Diabetmin 250mg Tablet only
If necessary, medication can be increased gradually to a maximum of 2.5g daily.

If transfer from another oral antidiabetic agent is intended; discontinue the other agent and initiate metformin at the dose indicated above.

Combination with insulin:
Metformin and insulin may be used in combination therapy to achieve better blood glucose control. Metformin is given at the usual starting dose of one tablet 2-3 times daily while insulin dosage is adjusted on the basis of blood glucose measurements.

Usual children dose
Metformin is not recommended for use in children.

Usual geriatric dose
Please refer to adult dose.
(Due to potential for decreased renal function, the dosage should be adjusted based on renal function and maximum doses are not advised for use in the elderly.)

Renal impairment
A GFR should be assessed before initiation of treatment with metformin containing products and at least annually thereafter. In patients at an increased risk of further progression of renal impairment and in the elderly, renal function should be assessed more frequently, e.g. every 3-6 months.

GFR mL/min	Total maximum daily dose (to be divided into 2-3 daily doses)	Additional considerations.
60-89	3000 mg	Dose reduction may be considered in relation to declining renal function.
45-59	2000 mg	Factors that may increase the risk of lactic acidosis should be reviewed before considering initiation of metformin. The starting dose is at most half of the maximum dose.
30-44	1000 mg	
<30	-	Metformin is contraindicated.

The text "to be divided into 2-3 daily doses" should be omitted for extended release products containing metformin as single agent.

Note: The information given here is limited. For further information, kindly consult your doctor or pharmacist.

Storage: Store below 30°C. Protect from light and moisture.

Presentation/Packing: Blister pack of 10 x 10's

Product Registration Holder: HOVID Bhd.
121, Jalan Tunku Abdul Rahman (Jalan Kuala Kangsar),
30010 Ipoh, Perak, Malaysia.

Manufactured by: HOVID Bhd.
Lot 56442, 7 1/2 Miles, Jalan Ipoh/Chemor,
31200 Chemor, Perak, Malaysia.

Revision date: April 2024